



Analog IC Design

CMOS Technology

MOHAMMED NASR EL-DIN

Ex 9.10 Section 9.4.1 Johns-Martin

Example 9.10

Estimate the total Output noise rms value for a 10KHz low-pass filter, as shown in Fig. 9.18, when $C_f = 160pF$, $R_f = 100K$, $R_1 = 10K$, and $R_2 = 9.1K$. Also, find the SNR for an input signal equal to 100 mV rms. Assume that the noise voltage of the opamp is given by $V_n(f) = 20 nV/\sqrt{Hz}$, both its noise currents are $I_n(f) = 0.6 pA/\sqrt{Hz}$, and that its unity-gain frequency equals 5 MHz.

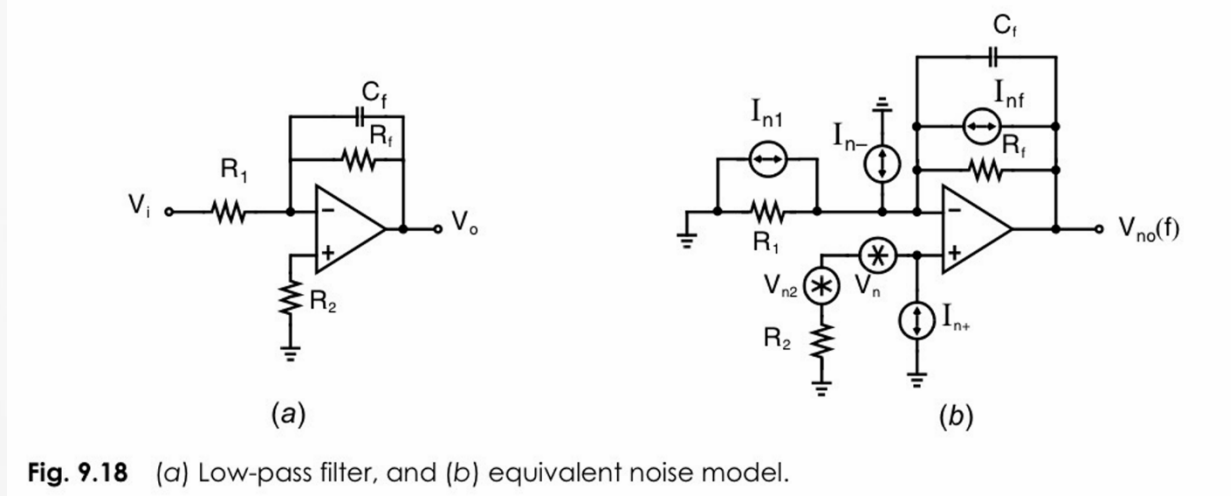


Fig. 9.18 (a) Low-pass filter, and (b) equivalent noise model.

Assume nominal temperature, $KT = 4.14 \times 10^{-21}J$

$$I_{n1}^2 = \frac{4KT}{R_1}, I_{nf}^2 = \frac{4KT}{R_f}, V_{n2}^2 = 4KTR_2$$

$$\Rightarrow I_{n1}^2 \approx (1.29 pA/\sqrt{Hz})^2$$

$$\Rightarrow I_{nf}^2 \approx (0.41 pA/\sqrt{Hz})^2$$

$$\Rightarrow V_{n2}^2 \approx (12.28 nV/\sqrt{Hz})^2$$

Using superposition and assuming all noise sources are uncorrelated.

Let V_{no1}^2 is the output noise density due to the inverting terminal noise sources “inverting amplifier” I_{n1} , I_{nf} and I_{n-}

The sum of the three currents will feed the feedback impedance $R_f // \frac{-j}{2\pi f C_f}$

$$\Rightarrow V_{no1}^2(f) = [I_{n1}^2 + I_{nf}^2 + I_{n-}^2] \left| \frac{R_f}{1 + j 2\pi f R_f C_f} \right|^2 = [I_{n1}^2 + I_{nf}^2 + I_{n-}^2] R_f^2 \times \frac{1}{1 + (2\pi f R_f C_f)^2}$$

This equation indicates that this part of the output noise = $[I_{n1}^2 + I_{nf}^2 + I_{n-}^2] R_f^2$ and shaped by the LPF 3dB

$$\text{frequency } f_o = \frac{1}{2\pi R_f C_f} \Rightarrow B_{N1} = \frac{\pi}{2} f_o = \frac{1}{4R_f C_f}$$

$$\Rightarrow V_{no1-rms}^2 = V_{no1}^2(0) \times \frac{\pi}{2} f_o = V_{no1}^2(0) \times \frac{1}{4R_f C_f}$$

Let V_{no2}^2 is the output noise density due to the non-inverting terminal noise sources “non-inverting amplifier” I_{n+} , V_n and V_{n2}

Converting I_{n+} to voltage source by multiplying it by R_2

$$\Rightarrow V_{no2}^2(f) = [I_{n+}^2 R_2^2 + V_n^2 + V_{n2}^2] \left| 1 + \frac{R_f/R_1}{1+j 2\pi f R_f C_f} \right|^2$$

This equation indicates that this part of the output noise = $[I_{n+}^2 R_2^2 + V_n^2 + V_{n2}^2]$ and shaped by the transfer function $\left| 1 + \frac{R_f/R_1}{1+j 2\pi f R_f C_f} \right|^2$

If we say $R_f \gg R_1 \Rightarrow R_f/R_1 \ll 1$

At low frequency, the gain roughly $\approx \frac{R_f}{R_1}$, and shaped by the 3dB frequency $f_o = \frac{1}{2\pi R_f C_f}$

Above f_o , the gain decreases by $-20dB/decade$ and reach unity around $f_1 = \frac{R_f}{R_1} f_o$

At high frequency, the gain decrease to unity and remains at this level

$$\Rightarrow V_{no2-rms}^2 = V_{no2}^2(0) \times \frac{1}{4R_f C_f} + [I_{n+}^2 R_2^2 + V_n^2 + V_{n2}^2] \times \frac{\pi}{2} (f_u - f_1), \text{ where } f_u \text{ is the unity gain frequency}$$

$$\Rightarrow V_{no}^2 = V_{no1}^2 + V_{no2}^2 \text{ or } V_{no-rms}^2 = V_{no1-rms}^2 + V_{no2-rms}^2$$

$$V_{no1}^2(0) = [I_{n1}^2 + I_{nf}^2 + I_{n-}^2] R_f^2 = \left[\left(1.29 \frac{pA}{\sqrt{Hz}} \right)^2 + \left(0.41 \frac{pA}{\sqrt{Hz}} \right)^2 + \left(0.6 \frac{pA}{\sqrt{Hz}} \right)^2 \right] (100K)^2 \approx (148nV/\sqrt{Hz})^2$$

$$\Rightarrow V_{no1-rms}^2 = (148nV/\sqrt{Hz})^2 \times \frac{1}{4 \times 100K\Omega \ 160pF} \approx (18.5\mu V)^2$$

$$V_{no2}^2(0) = [I_{n+}^2 R_2^2 + V_n^2 + V_{n2}^2] \times \left(1 + \frac{R_f}{R_1} \right)^2 = \left[\left(0.6 \frac{pA}{\sqrt{Hz}} \ 9.1K\Omega \right)^2 + \left(20 \frac{nV}{\sqrt{Hz}} \right)^2 + \left(12.28 \frac{nV}{\sqrt{Hz}} \right)^2 \right] \times \left(1 + \frac{100K\Omega}{10K\Omega} \right)^2$$

$$\approx \left(24.1 \frac{nV}{\sqrt{Hz}} \right)^2 \times 11^2 = \left(265 \frac{nV}{\sqrt{Hz}} \right)^2$$

$$\Rightarrow V_{no2-rms}^2 = \left(265 \frac{nV}{\sqrt{Hz}} \right)^2 \times \frac{1}{4 \times 100K\Omega \ 160pF} + \left[\left(0.6 \frac{pA}{\sqrt{Hz}} \ 9.1K\Omega \right)^2 + \left(20 \frac{nV}{\sqrt{Hz}} \right)^2 + \left(12.28 \frac{nV}{\sqrt{Hz}} \right)^2 \right] \times \frac{\pi}{2} (f_u - f_1)$$

$$\approx (74.6\mu V)^2, \text{ where } (f_u - f_1) = 5MHz - \frac{100K\Omega}{10K\Omega} \frac{1}{4 \times 100K\Omega \ 160pF} \approx 4.84MHz$$

$$\Rightarrow V_{no-rms}^2 = (18.5\mu V)^2 + (74.6\mu V)^2 \approx (76.86\mu V)^2 \text{ or } V_{no-rms} = 76.86\mu V$$

$$\text{The output signal level} = \text{input signal level} \times \frac{R_f}{R_1} = 100mV_{rms} \times \frac{100K\Omega}{10K\Omega} = 1V_{rms}$$

$$\Rightarrow SNR = 10 \log \left(\frac{V_o^2}{V_{no-rms}^2} \right) = 10 \log \left(\frac{1V^2}{(76.86\mu V)^2} \right) \approx 82.3dB$$